

Stuck & Recovery Analysis

Endovascular navigation RL · run v1b · explore-only

14,691

episodes

4,055,067

explore steps

58.6%

episode success

13,013

stall events

SCOPE. Every number in this report is measured from the v1b worker-log complete ledger (EPISODE_START / STEP / EPISODE_OUTCOME), restricted to EXPLORE episodes. Evaluation episodes are excluded exactly: runner.eval() resets with explicit seeds, so eval EPISODE_START lines carry a `seed=` field and explore resets do not. That filter yields 14,691 explore episodes and exactly 1,470 eval (15 evals x 98) — exact. The obvious alternative, main.log time windows, leaks; and episode_summary.jsonl is unusable because it logs only ~36% of explore episodes. TRAINING-TIME AXIS. Cumulative explore steps, summed per-episode in chronological order—exact from the logs, unlike wall-clock (which conflates simulator speed) or the global step counter. THIS RUN. v1b is the P2-teacher residual-on-heuristic baseline. SELF-CONTAINED. Every table and every claim in this report is derived from this run alone. Nothing here depends on comparison with another run.

Definitions — and why each one is drawn where it is

Every downstream number depends on these choices, so they are stated before any result

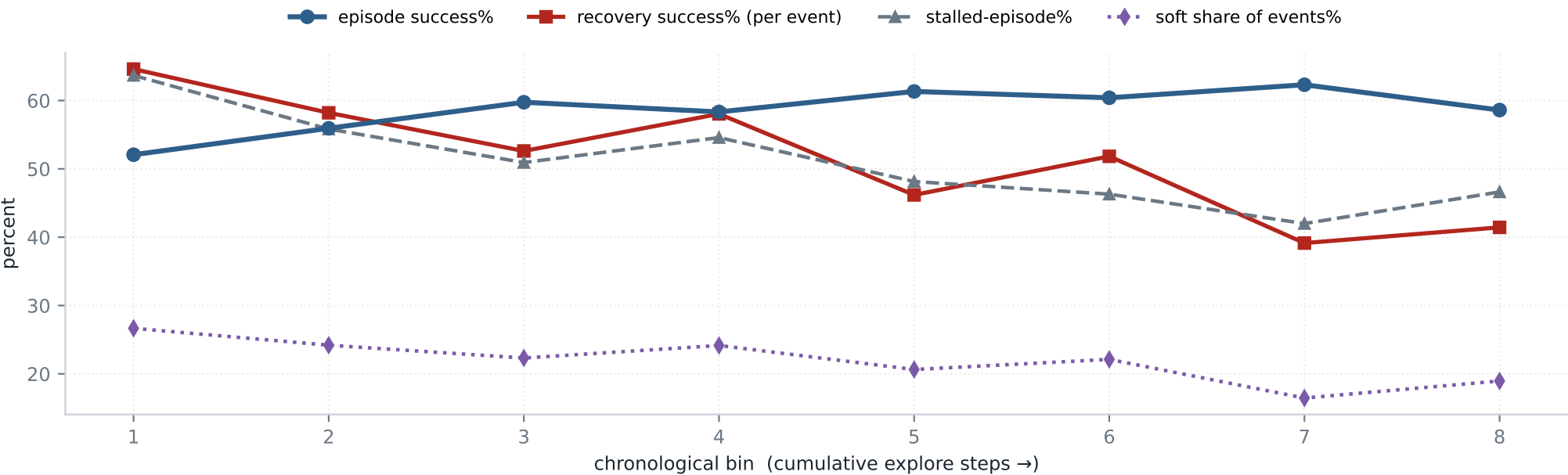
term	unit	definition
stall event	per event	wire pushing ($ gw_trans > 2$ mm/s) while path progress stays flat (< 0.3 mm gain) for 12 consecutive steps
grind	per event	escaped the stall after < 1 mm of withdrawal — forced through
soft	per event	escaped after 1–8 mm withdrawal — ease slack, re-advance (the manoeuvre we set out to induce)
hard	per event	escaped after > 8 mm withdrawal — costly pullback
unrecovered	per event	never passed the jam; terminal for that episode
recovery success%	per EVENT	(soft + hard + grind) / all stall events
episode success%	per EPISODE	reached the target

A stall is 'pushing but not progressing', not merely 'not progressing'. A wire that is deliberately retracting or rotating is manoeuvring, not stuck. Requiring the push command isolates the case where the policy IS trying to advance and the vessel will not allow it. The stall counter decays by 2 per non-stalled step rather than resetting, so one noisy command sign-flip inside a genuinely stuck push does not erase the evidence. Recovery kind is classified by ACTUAL withdrawn length — peak insertion minus deepest insertion during the stall — not by the commanded retraction. A retract command that moves nothing (wire wedged) must not score as a soft recovery. THE THREE SUCCESSES sit at different units of analysis, which is the single most important thing on this page: (1) EPISODE success is per episode — did it reach target. (2) RECOVERY success is per EVENT — did the wire escape a given stall. (3) RECOVERY-TO-EPISODE success is per episode — did escaping actually convert into reaching the target. Conflating (1) and (2) makes 'success tracks recovery' look true when the opposite can hold.

Stuck & recovery vs training time

Each bin ~ 1800 episodes of training data

bin	explore steps (k)	success %	stalled-ep %	events	recovery %	soft %	hard %	grind %	unrecovered %
1	0-603	52%	64%	2441	65%	27%	11%	27%	35%
2	604-1148	56%	56%	1926	58%	24%	14%	20%	42%
3	1149-1647	60%	51%	1568	53%	22%	12%	18%	47%
4	1647-2157	58%	55%	1797	58%	24%	12%	21%	42%
5	2157-2629	61%	48%	1323	46%	21%	8%	17%	54%
6	2629-3108	60%	46%	1501	52%	22%	11%	18%	48%
7	3108-3559	62%	42%	1142	39%	16%	10%	13%	61%
8	3559-4053	59%	47%	1308	41%	19%	6%	17%	59%



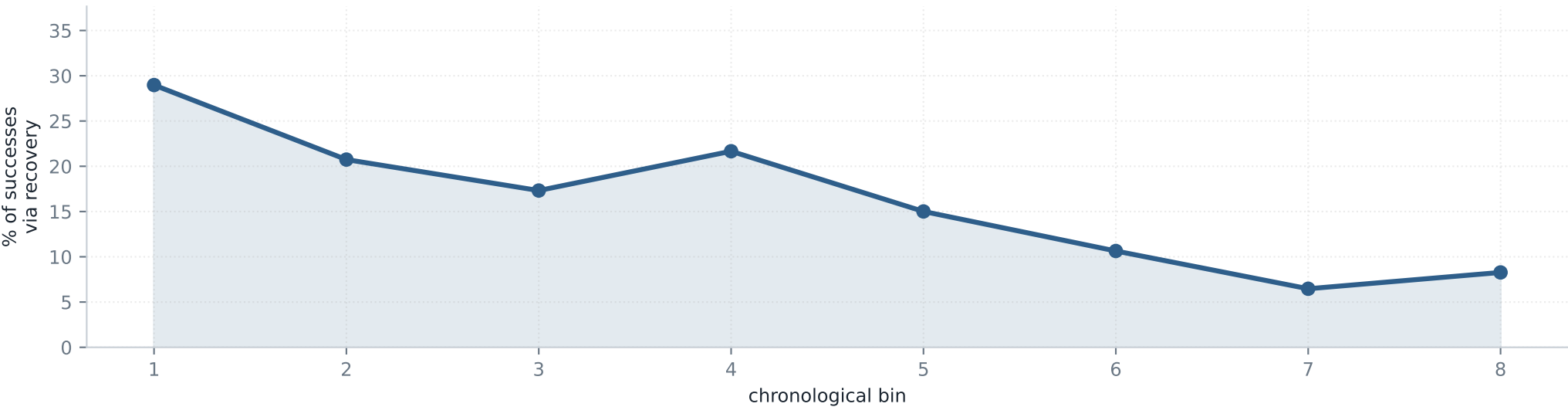
WHAT IT SHOWS Fig 1: Success rates and recovery rates with training Fig 2. Falling recovery rates creates a ceiling for success rate

INFERENCE Episode success CLIMBS 52→62% while recovery success FALLS 65→39% and unrecovered rises. What is actually improving is the stalled-episode share, which drops 64→42%. The policy is not learning to escape stalls; it is learning not to enter them, and the escape skill withers as stalls leave its training data.

Success vs Recovery

Each episode assigned to exactly one channel: clean / stuck-and-recovered / stuck-and-failed

bin	success %	clean eps (success %)	recovered eps (success %)	unrec eps (success %)	% of successes via recovery
1	52%	667 (100%)	305 (91%)	864 (1%)	29%
2	56%	811 (100%)	220 (97%)	805 (0%)	21%
3	60%	901 (100%)	192 (99%)	743 (1%)	17%
4	58%	834 (100%)	248 (94%)	754 (1%)	22%
5	61%	952 (100%)	172 (98%)	712 (1%)	15%
6	60%	986 (100%)	127 (93%)	723 (1%)	11%
7	62%	1065 (100%)	76 (97%)	695 (1%)	6%
8	59%	980 (100%)	90 (99%)	766 (1%)	8%



WHAT IT SHOWS Episode success decomposed by channel, and the share of all successes that flowed through recovery (chart).

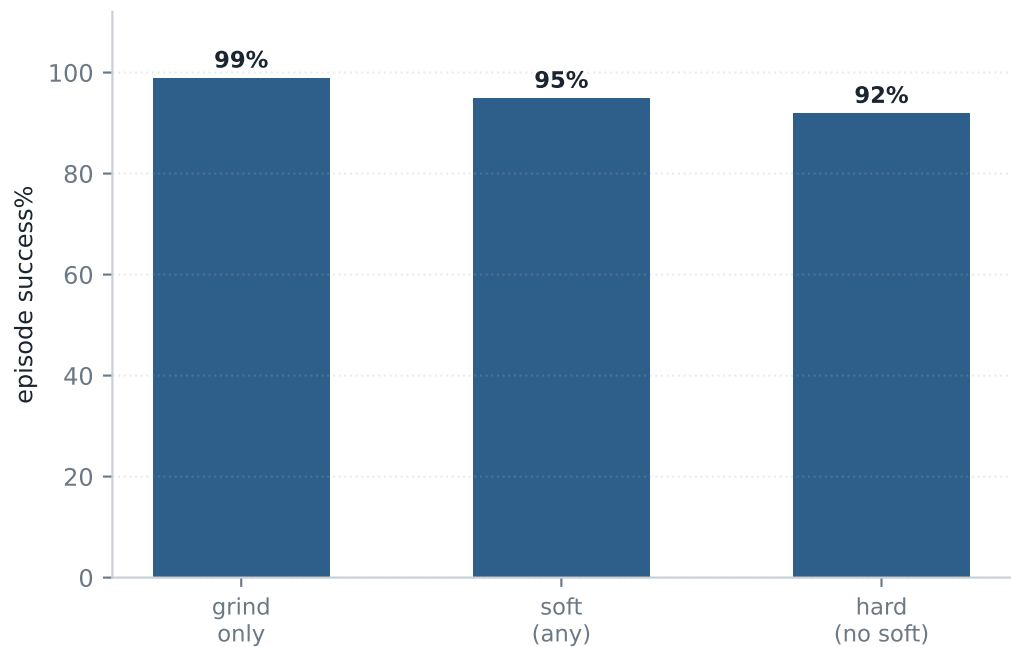
***INFERENCE** Recovery's contribution COLLAPSES from 29% to 6%. Early on roughly a third of all wins were rescues; by the end nearly every win is an episode that simply never jammed (clean episodes rise 305→76). Note: As seen by the success rates in columns 4 and 5, a small fraction of episodes that recover do not reach the target as they have exhausted all their steps and similarly some that do get stuck are already very close to the target and hence gets counted as successes.*

Recovery type vs episode success

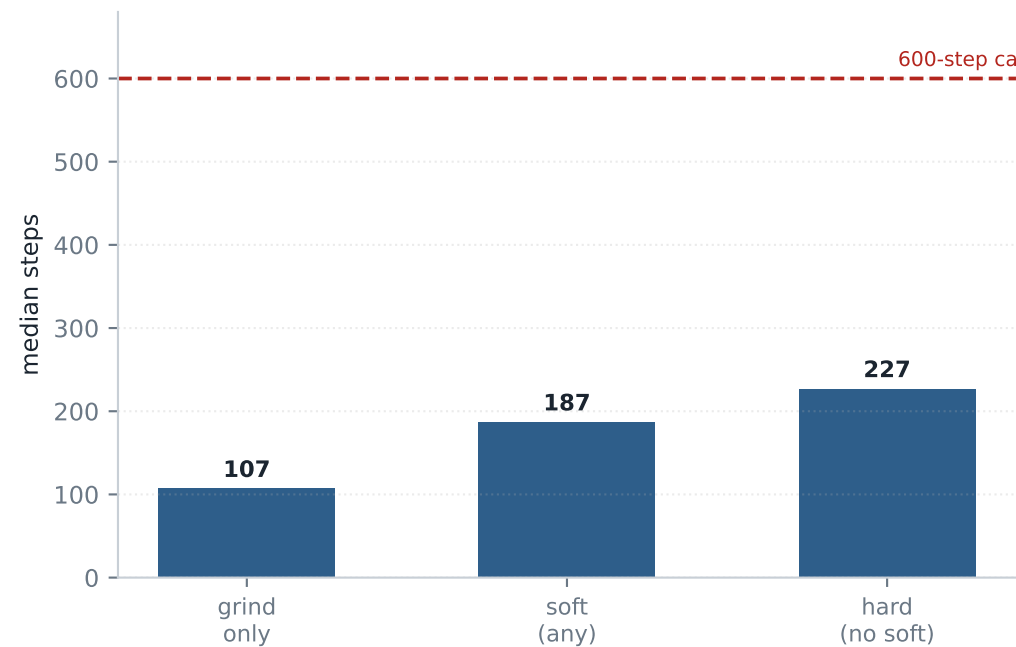
Episodes grouped by the recovery they achieved

recovery profile	success %	median steps	mean retract mm
soft (any)	95%	187	4.9
hard (no soft)	92%	227	13.7
grind only	99%	107	0.1
unrecovered (any)	1%	600	NA
never stalled	100%	41	0.0

episode success by recovery type



time cost of each recovery type



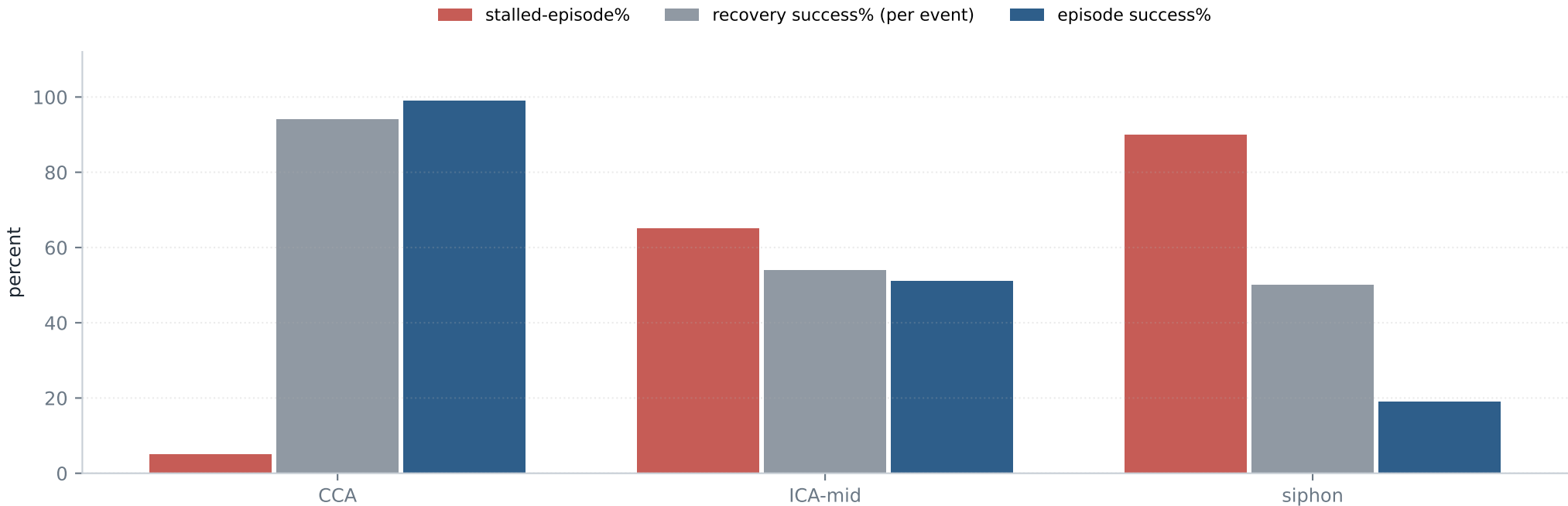
WHAT IT SHOWS Whether the type of escape predicts the episode outcome, and what each type costs in steps.

INFERENCE GRIND BEATS SOFT: grind-only episodes succeed 99% in 107 median steps, against soft at 95% in 187 steps and hard at 92% in 227. The likely reason is a confound: recovery type is mostly a proxy for stall SEVERITY, not policy skill. A jam you can grind through was mild; one forcing an 8 mm withdrawal was severe and usually deeper, where the episode was already in trouble.

Stuck & recovery by depth

Path-length bands: CCA (<146 mm) · ICA-mid (146–210 mm) · siphon (>210 mm)

band	eps	ep succ%	stalled-ep%	events	recovery succ%	soft%	hard%	grind%	unrec%
CCA	5368	99%	5%	496	94%	43%	15%	36%	6%
ICA-mid	4692	51%	65%	5063	54%	23%	14%	18%	46%
siphon	4631	19%	90%	7454	50%	21%	9%	20%	50%

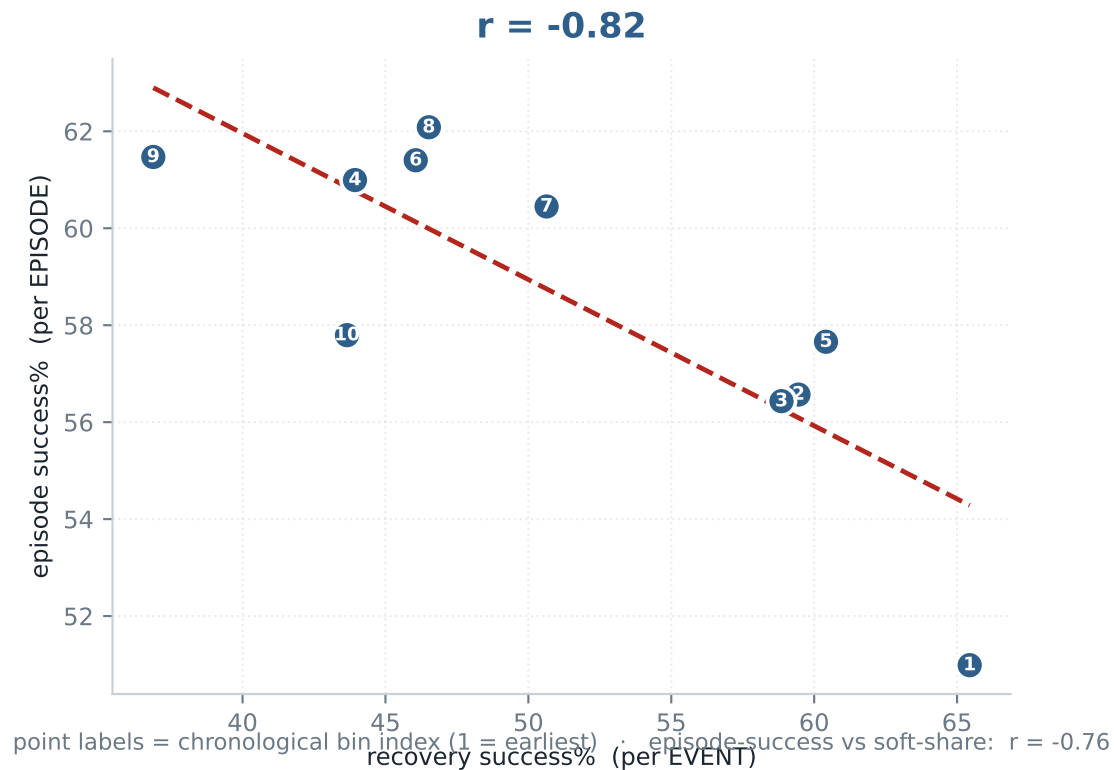


WHAT IT SHOWS Where the stalls actually are, and whether escaping them rescues the episode there.

INFERENCE Depth dominates every other factor. Stalling runs 5% at the CCA but 90% at the siphon — getting stuck is essentially guaranteed once the wire enters the cavernous segment. And escaping does not save you there: 50% of siphon stalls are escaped yet only 19% of siphon episodes succeed. You escape one jam and immediately meet the next. This reframes the siphon as a STALL-DENSITY problem rather than an escape-skill problem — the same wall the real-patient evaluation hit at 0 of 30, seen from the inside.

Is episode success actually tracking recovery?

10 chronological bins · each point is one bin · units deliberately mixed to test the intuition



WHAT IT SHOWS A direct test of the intuition that success rises and falls with recovery skill. Each point is one chronological bin; the dashed line is the least-squares fit.

INFERENCE **STRONGLY NEGATIVE** ($r = -0.82$). Periods of better recovery are periods of **WORSE** success, because both are downstream of how often the wire is jamming at all. An earlier reading in this program that 'success% $\approx$ recovery%' was wrong — it compared a per-EVENT rate against a per-EPIISODE rate without noticing they are different units, which is precisely why the three successes are separated in T2.

Threshold sensitivity — are these conclusions detector artifacts?

Every episode scored under three detector configurations simultaneously

v1b under three detectors

config	events	stalled-ep%	recovery succ%	soft%	hard%	grind%	unrec%
sens	17,386	55%	65%	29%	10%	26%	35%
canon	13,013	51%	53%	23%	11%	20%	47%
strict	10,394	48%	42%	18%	11%	14%	58%

detector parameters

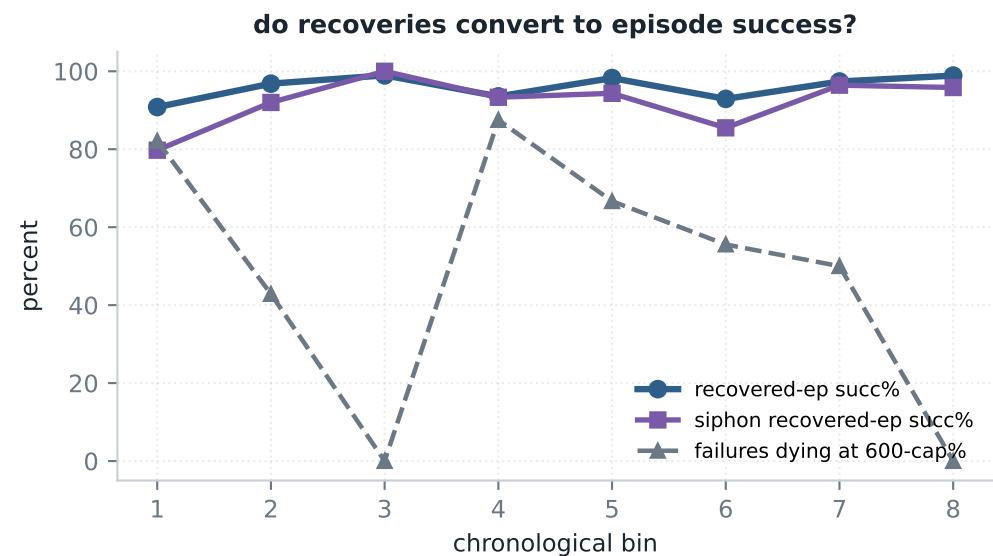
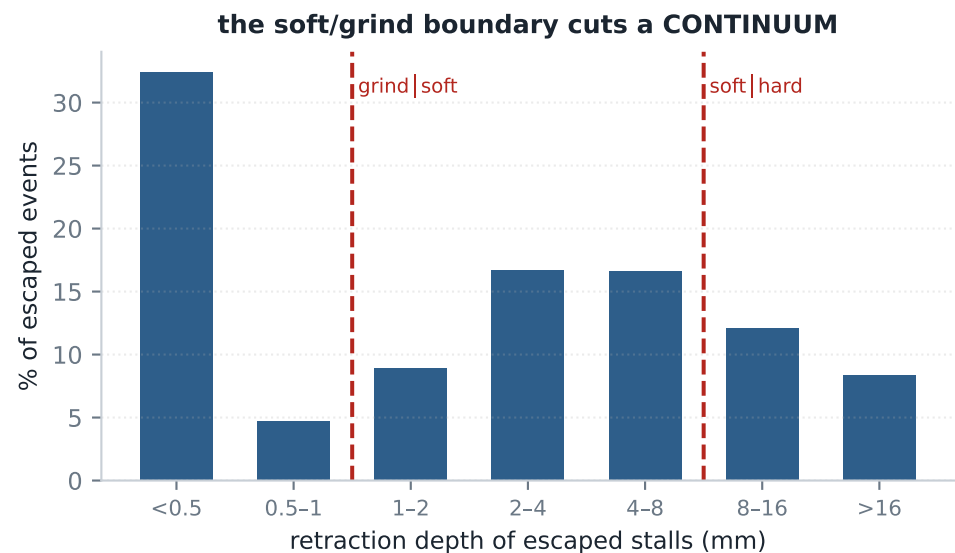
config	stall_eps	push_min	stuck_steps	retract_min	soft_max	pass_eps
sens	0.3 mm	1.0 mm/s	8	0.5 mm	8 mm	1.0 mm
canon	0.3 mm	2.0 mm/s	12	1.0 mm	8 mm	1.0 mm
strict	0.5 mm	4.0 mm/s	16	1.0 mm	8 mm	1.0 mm

WHAT IT SHOWS The same analysis re-run under a sensitive, a canonical and a strict definition of 'stuck', to separate signal from detector choice.

***INFERENCE** Absolute rates move a great deal — recovery success reads anywhere from 42% to 65% for this run alone depending on strictness — so NO absolute number in this report should be quoted on its own. What survives all three settings is the ORDERING and the DIRECTION of every trend, which is why each finding here is stated as a delta or a trend rather than a level.*

Diagnostics — recovery quality, cost and where it fails

Retraction-depth distribution, per-bin conversion, and the forensics of recovered-but-failed episodes



conversion of recovered episodes, by depth

band	recovered eps	episode succ%
CCA	251	98%
ICA-mid	733	97%
siphon	446	91%

what the failures were doing

recovered-but-FAILED episodes	value
count	68 of 1430 recovered
died at the 600-step cap	71%
median final distance to target	34 mm

WHAT IT SHOWS Left: the retraction-depth distribution of escaped stalls, with the classification cuts drawn on. Right: how well this run's recoveries convert to episode success over training. Below: conversion by depth band and what the failed recoveries were doing when they ended.

INFERENCE The retraction distribution is a CONTINUUM massed near zero (32% of escapes are under 0.5 mm), not a bimodal soft/grind split — so the 1 mm and 8 mm cuts slice a smooth curve and absolute soft% is detector-defined rather than a property of the policy. Recovered episodes convert at 95% overall and the conversion is high in every band, so in this run a recovery that happens is a recovery that pays.

Findings

Every claim is stated as a delta or trend, and survives all three detector configurations

F1 Improvement is stall AVOIDANCE, not stall ESCAPE.

Episode success rises 52→62% while recovery success falls 65→39%, unrecovered climbs, and recovery's share of successes collapses 29→6%. The correlation is $r = -0.82$. The policy is not learning to get unstuck; it is learning not to get stuck, and the recovery skill decays as it does. This is the interference mechanism the design doc predicted, now measured.

F2 GRIND beats SOFT on episode success.

Grind-only episodes succeed 99% in 107 median steps; soft 95% in 187; hard 92% in 227. This inverts the premise behind the crunchpass lane. Likely a confound — recovery type proxies stall SEVERITY rather than policy skill — so soft recovery cannot be a training target until that is broken.

F3 Recovery, when it happens, pays here.

Recovered episodes convert at 95% and stay high in every depth band. The problem is not that this run's recoveries are bad; it is that they become RARE as avoidance takes over, so they stop contributing to the headline number.

F4 Depth dominates everything.

90% of siphon episodes stall versus 5% at CCA. At the siphon, escaping does not save the episode: 50% escape but only 19% succeed. The limiting factor is stall DENSITY, not escape ability — the same wall the real-patient eval hit at 0 of 30.

F5 Detector-robust.

Absolute rates move across the three configurations (recovery success 42-65%), but every trend direction holds under all of them.

Implications

What these measurements license, and what they rule out

1 The crunchpass lane's premise needs revisiting BEFORE launch.

It amplifies success-conditioned soft-ish passages on the theory that soft recovery is the skill to grow. F2 says grind-through converts better and F1 says the winning channel is avoidance. Break the severity confound first— compare soft vs grind within matched depth.

2 If recovery is wanted, it must be made to PAY.

Recovered episodes cost several times the steps of clean ones. Under a 600-step cap with a sparse target reward, a slow recovery is nearly worthless — the optimiser is CORRECT to prefer avoidance. Either the budget accommodates recovery or recovery gets cheaper.

3 The siphon wall is not a recovery problem.

Near-total stall rates with moderate escape but low success mean the limiting factor is the density of stalls, not the ability to escape any one. Interventions aimed at escape skill are aimed at the wrong target.

4 Beware the units trap.

The strong negative r is only visible once per-EVENT and per-EPISODE rates are separated. Any future analysis quoting a single 'recovery rate' against success is at risk of reproducing the earlier mistaken reading.

Reproduce: `monitoring/extract_stuck.py <v1b_run_dir> stuck_v1b.jsonl` then `monitoring/report_single.py stuck_v1b.jsonl v1b <out.pdf>`

All figures in this report are derived from this run alone.